

Winger Crystal

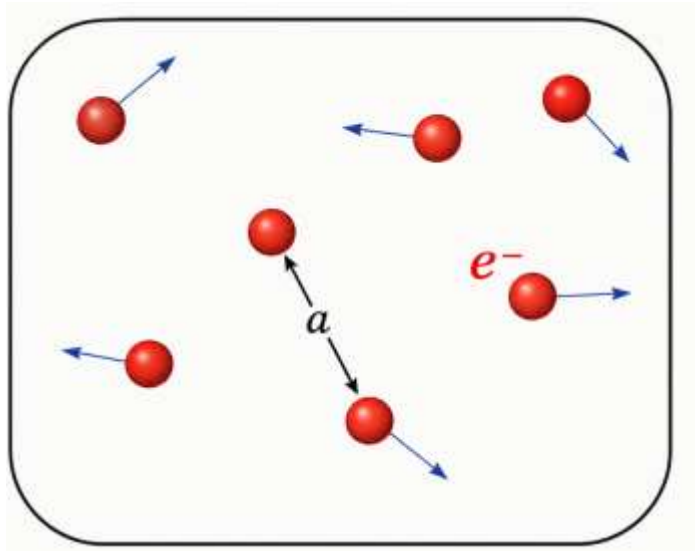
Sep 15th 2026

Outline

- Background
- Experimental realization
- Summary

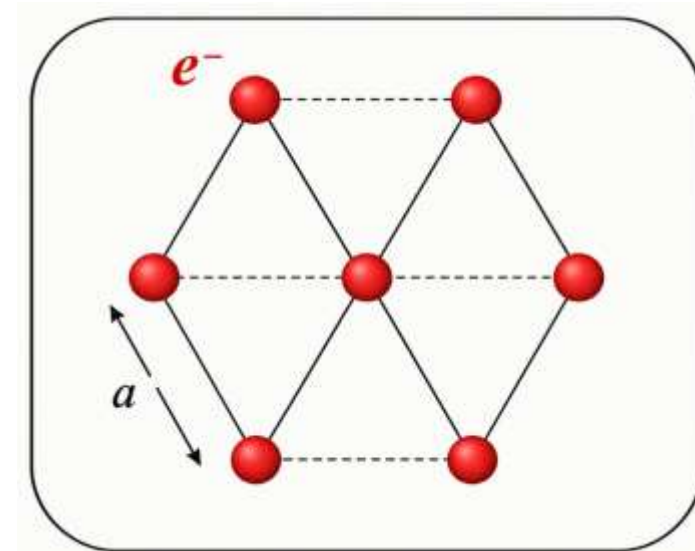
Background

2D electron gas



Gas, liquid

Kinetic energy dominate



Wigner crystal (Solid)

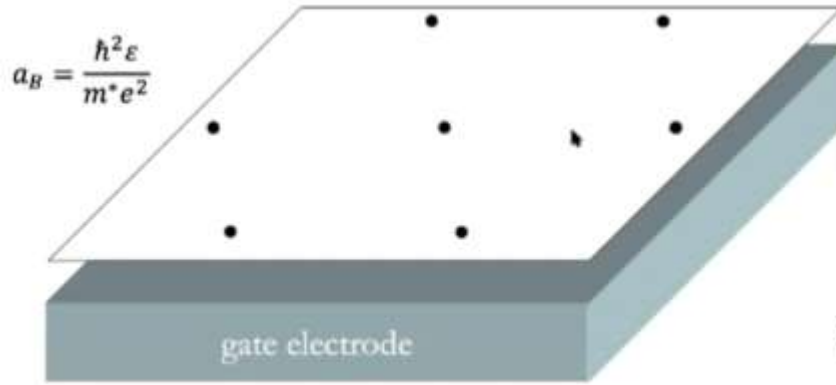
Potential energy dominate

Historical context of Wigner crystals:

1934: Theoretical prediction by Eugene Wigner .

1979: First indirect observation by C. Grimes and G. Adams.

2024: First imaging of Wigner crystals by Y. Tsui and colleagues.



$$\pi r_0^2 n_e = 1$$



$$r_0 = \frac{1}{\sqrt{\pi n_e}}$$

$$r_s = \frac{r_0}{a_B} = \frac{1}{\sqrt{\pi n_e a_B^2}}$$

Wigner-Seitz Radius r_s : Average distance between a pair of valence electrons in a uniform electron gas.

Conditions for Wigner crystallization: $E_C(r_s) > E_k(r_s)$

Coulomb Potential Energy $E_C(r_s)$

$$E_C(r_s) = \frac{e^2}{4\pi\epsilon r_s a_B} \propto \frac{1}{r_s}$$

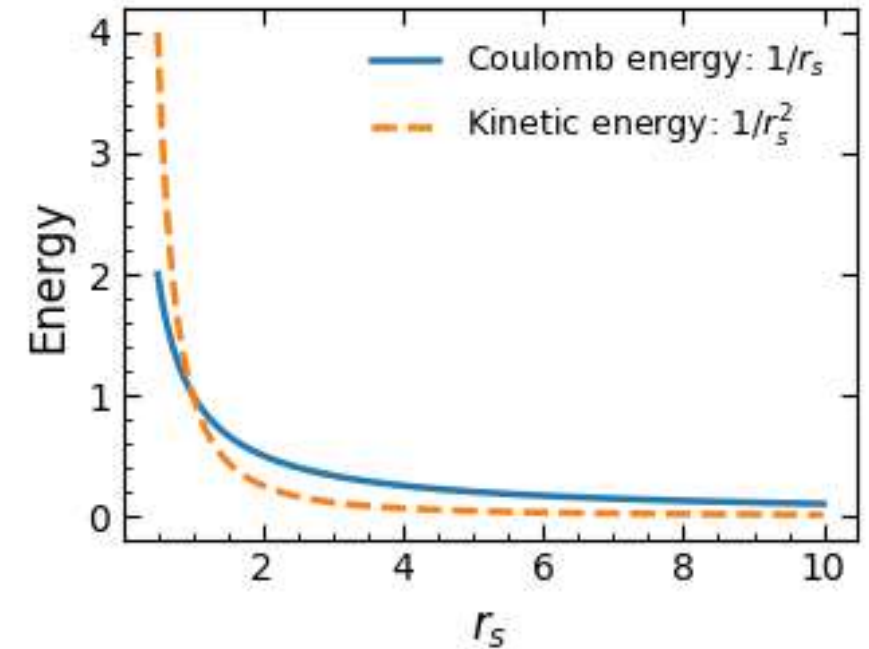
Kinetic energy $E_k(r_s)$

$$E_k(r_s) = \frac{\hbar^2}{m r_s^2 a_B^2} \propto \frac{1}{r_s^2}$$

Quantum

$$E_k(r_s) \propto k_B T$$

Classical (Finite temperature)



Wigner Crystals: A phase of electrons that occurs when the Coulomb potential between the electrons dominates over their kinetic energy. The electrons localize into a regular lattice structure to minimize their energy.

$$\Gamma = \frac{E_C}{E_k}$$

$$\Gamma < 1$$

Kinetic energy dominate

Electron gas

$$1 < \Gamma < 100$$

$$E_C \sim E_k$$

liquidlike

$$\Gamma > 100$$

Potential energy dominate

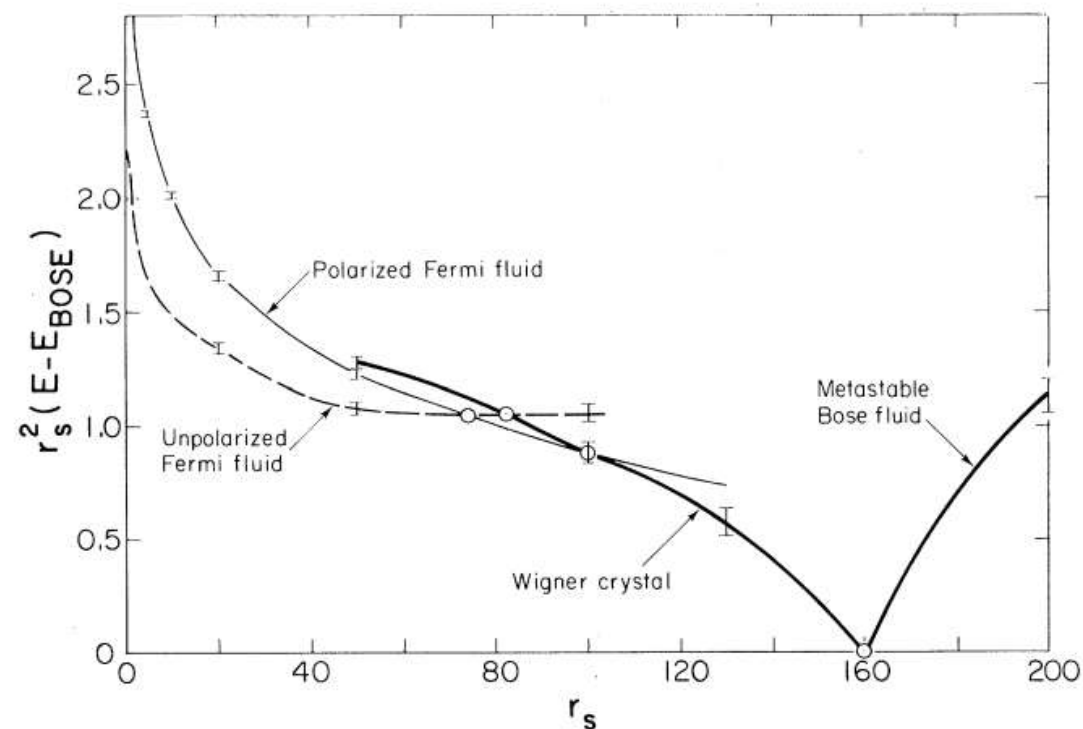
Electron Solid

Numerical method: Monte Carlo (MC)

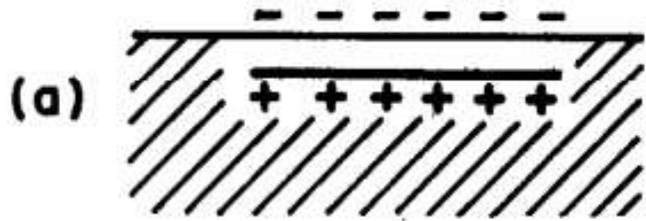
$$\frac{\partial f}{\partial t} = \frac{\hbar^2}{2m} \left[\sum_{i=1}^N \nabla_i^2 f - \nabla_i (f \nabla_i \ln |\Psi_T|^2) \right] - \left[\frac{H \Psi_T}{\Psi_T} - E_{\text{ref}} \right] f,$$

$$H = \left(\frac{\hbar^2}{2m} \right) \sum_{i=1}^N \nabla_i^2 - \sum_{i < j} e^2 / r_{ij}.$$

$$f(r, t) = \Psi_T \varphi_0 \exp[-t(E_{\text{ref}} - E_0)],$$

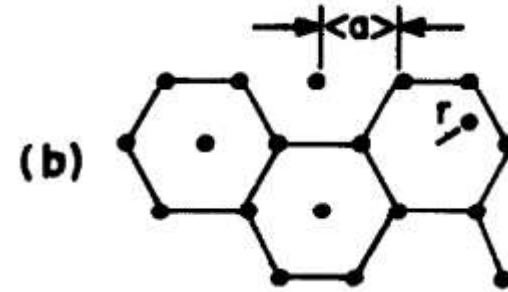


Experimental realization: Liquid helium

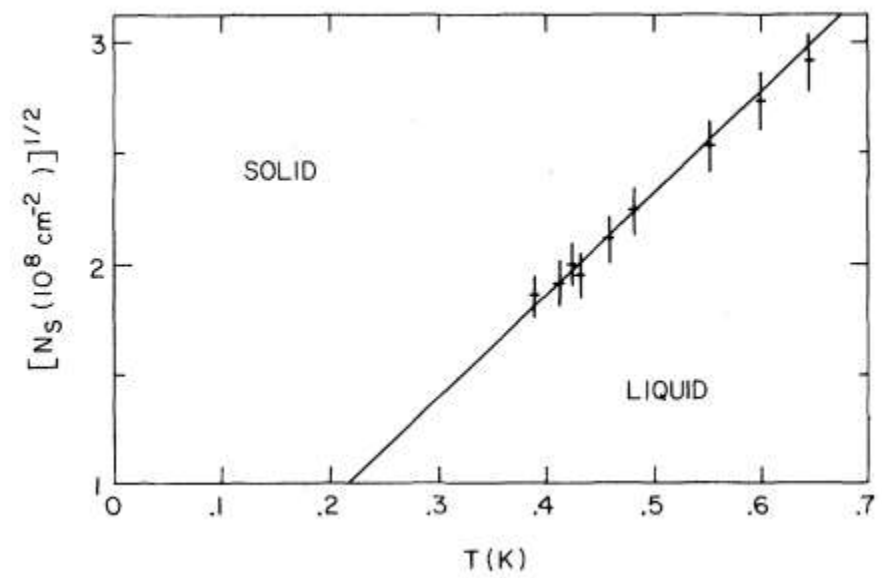
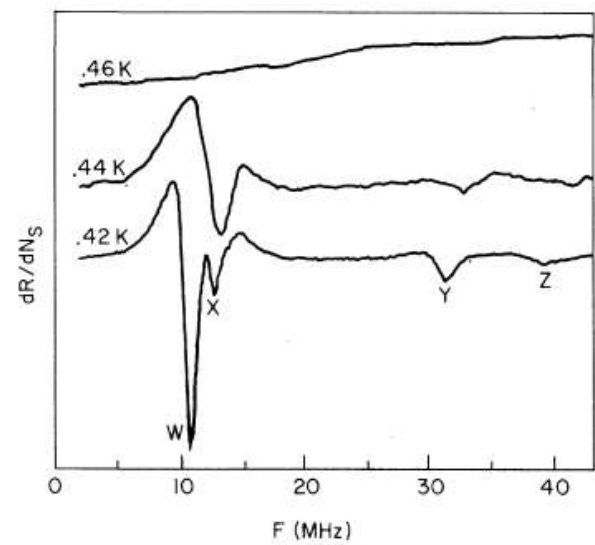


$$\omega_p^2 = 2\pi e^2 n_s q / \epsilon m.$$

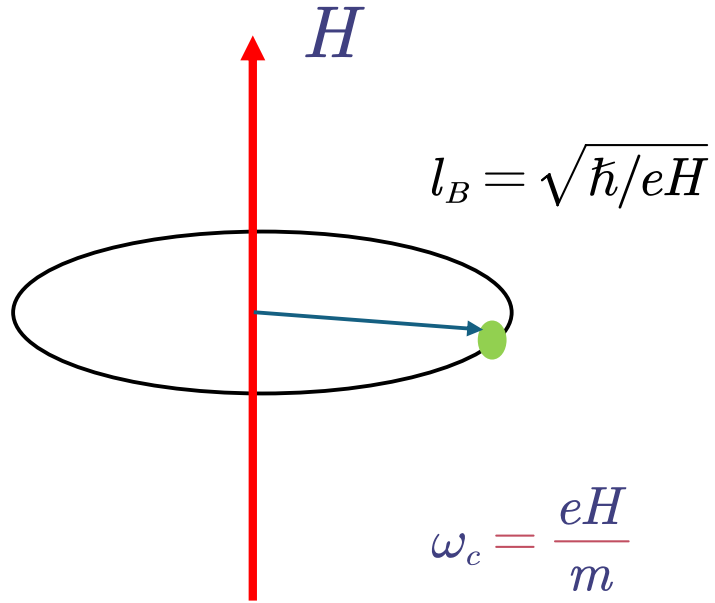
$$\Gamma = \frac{\pi^{1/2} N_s^{-1/2} e^2}{k_B T}$$



$$\nu_n = 2^{5/4} 3^{-3/8} \pi^{1/2} (\alpha/\rho)^{1/2} N_s^{3/4} n^{3/4}$$



Experimental realization: Quantum Hall system



Plasmon

$$\omega_p^2 = 2\pi e^2 n_s q / \epsilon m.$$

Add perpendicular magnetic field

$$\omega_+^2 = \omega_p^2 + \omega_c^2.$$

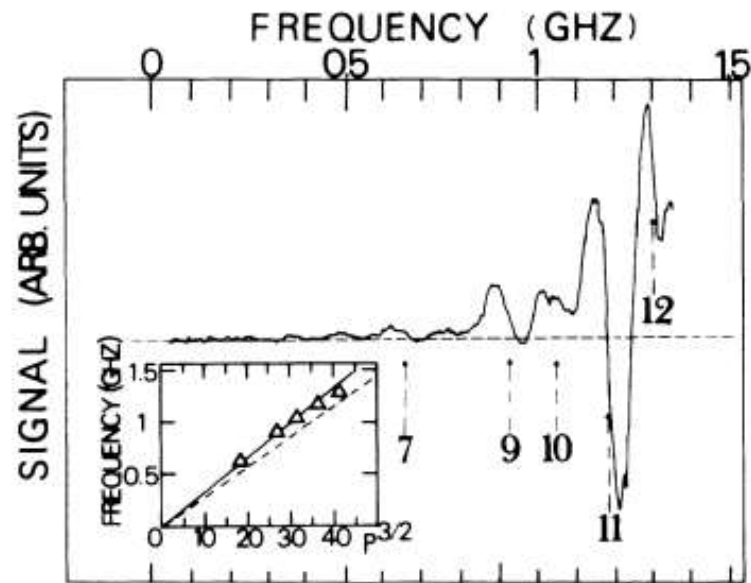
Gapped in bulk

magnetophonon branch

$$\omega_- = \omega_t^{(0)} \omega_p / \omega_c = c(2\pi\mu)^{1/2} q^{3/2} / H, \quad \omega_c \gg \omega_p,$$

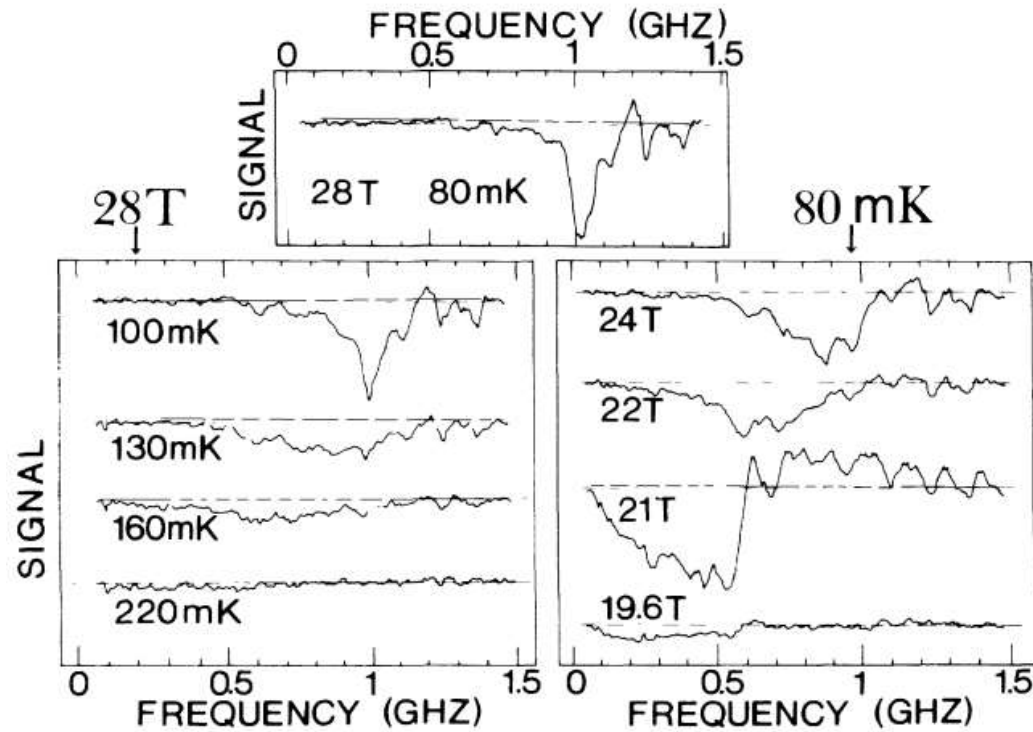
gapless

Experimental realization: Quantum Hall system

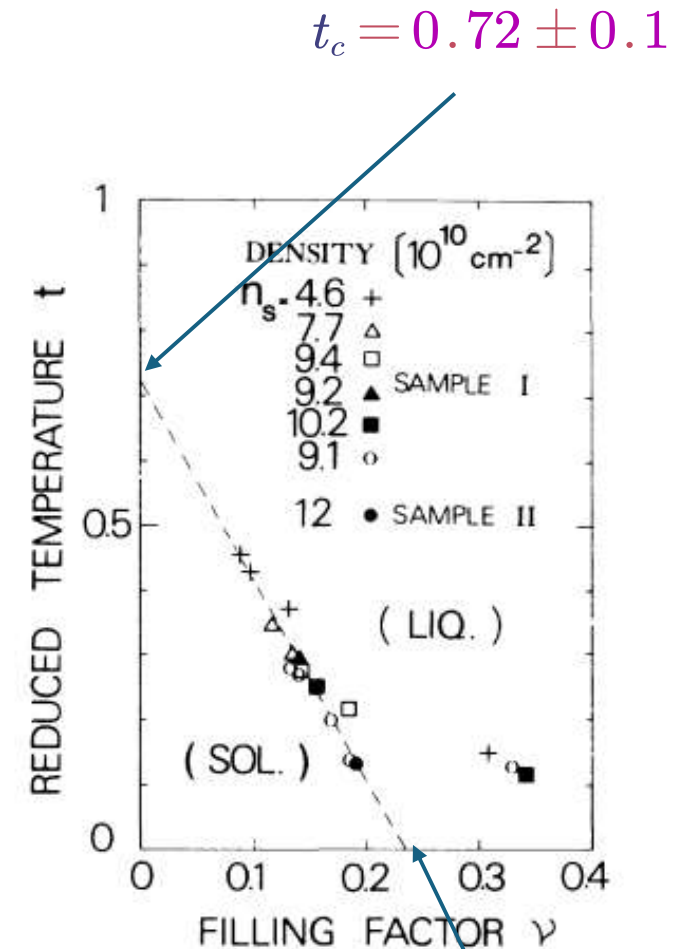


$$q = pq_0$$

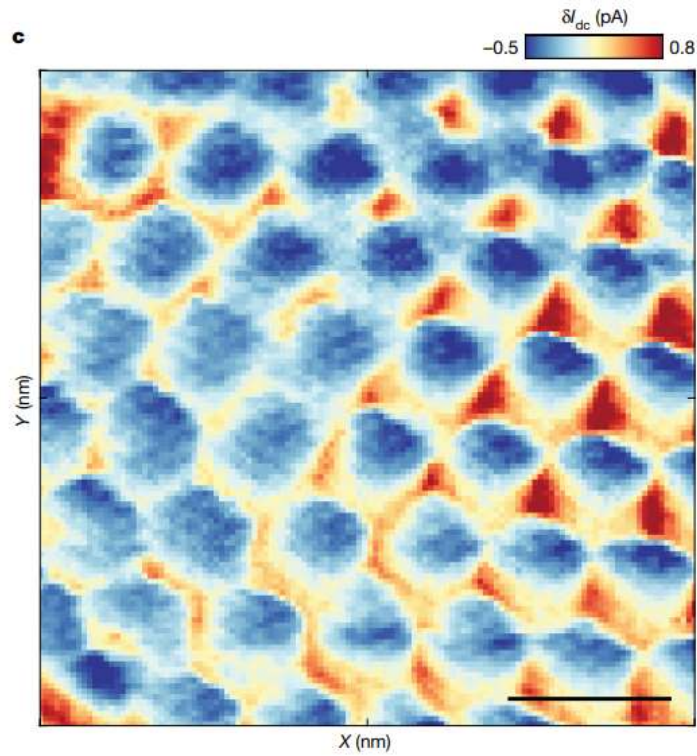
Experimental realization: Quantum Hall system



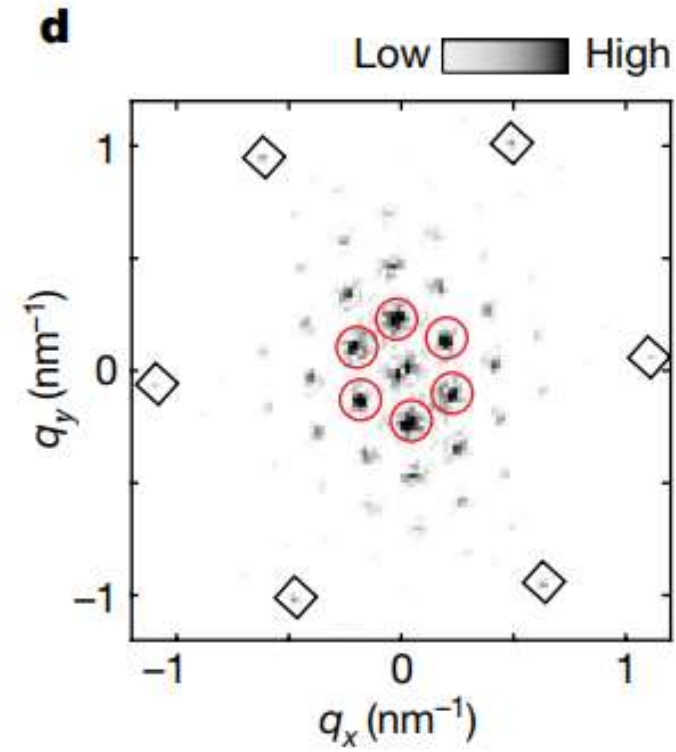
$$\nu = (3.92 \text{ T})/H,$$



Experimental realization: bilayer graphene

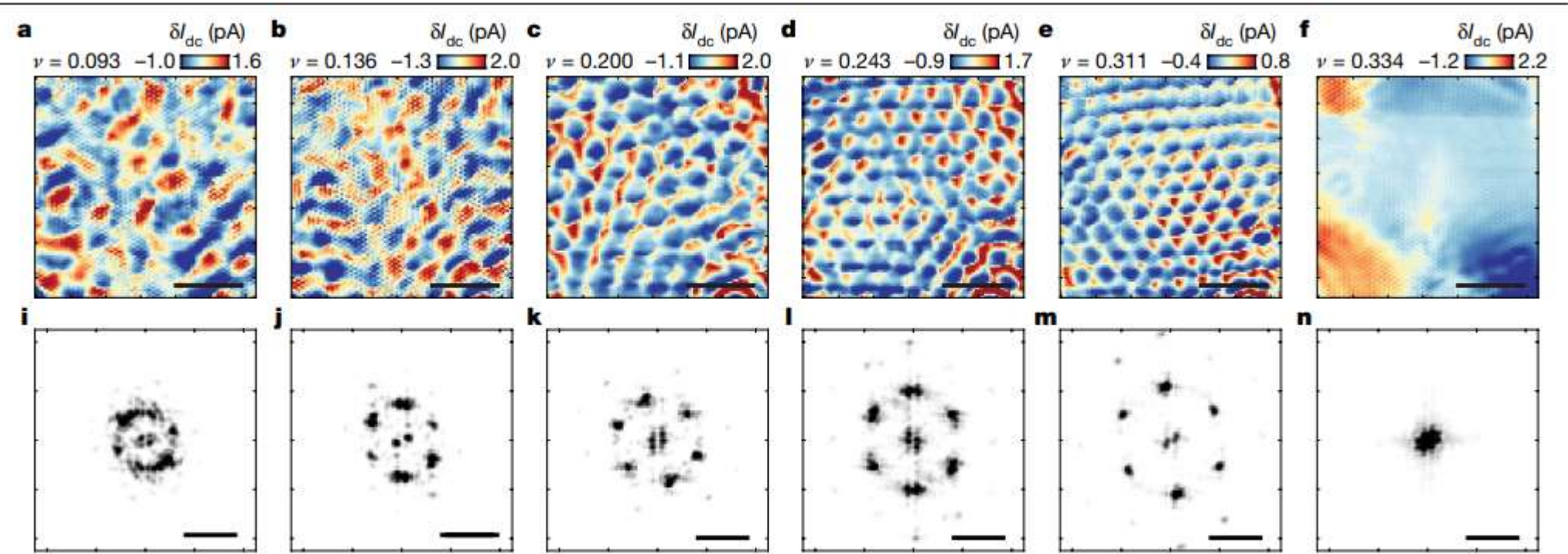


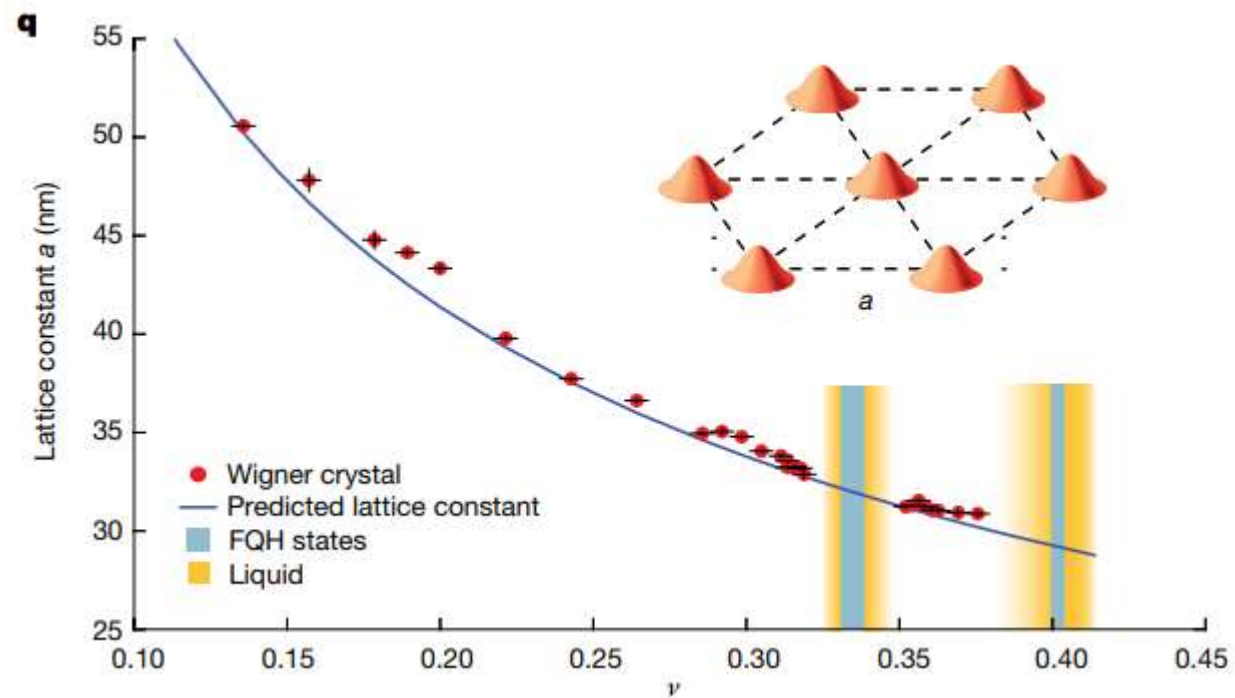
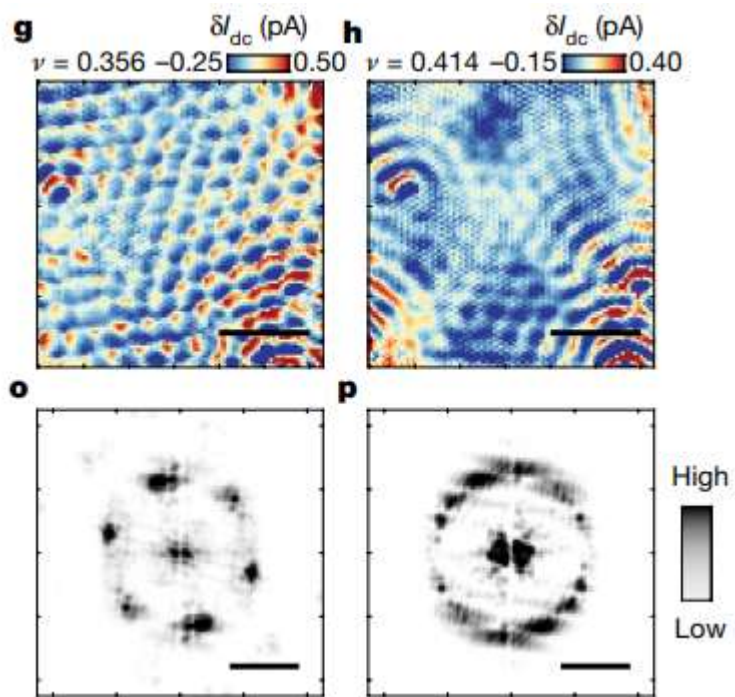
Real space



reciprocal-space

Phase diagram

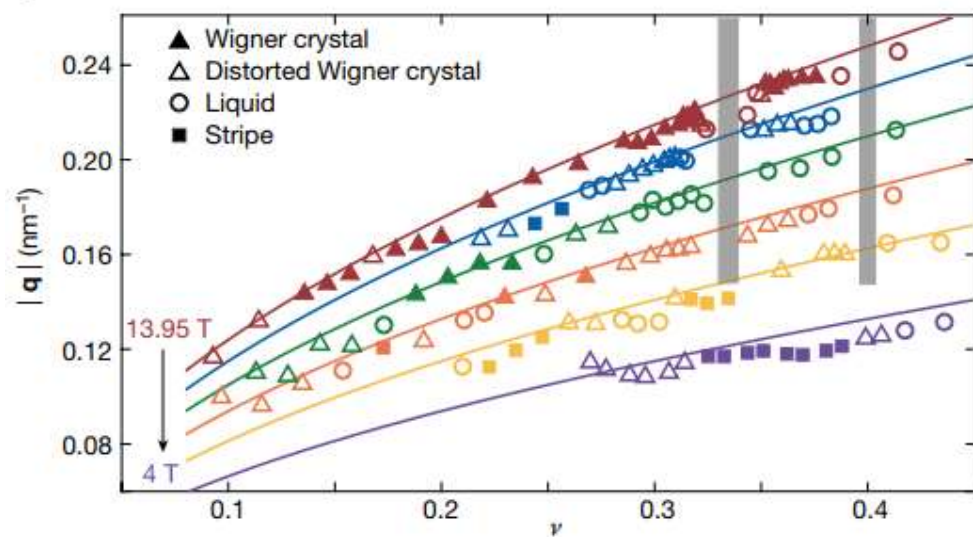




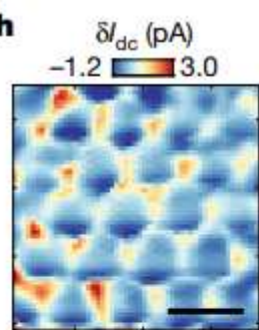
$$a = \sqrt{\frac{4\pi l_B^2}{\sqrt{3}v}}$$

stripe phase

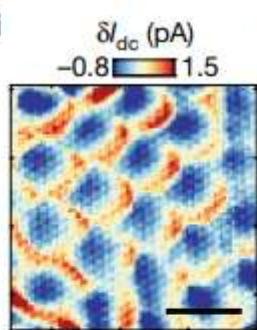
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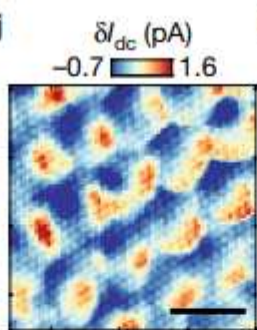
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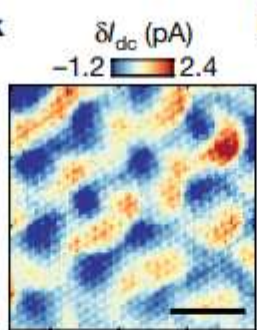
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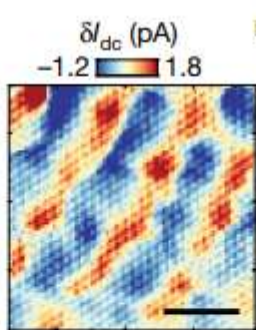
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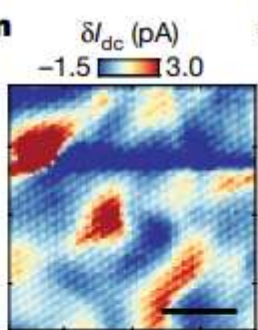
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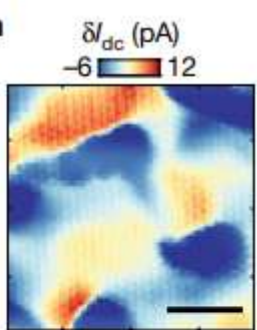
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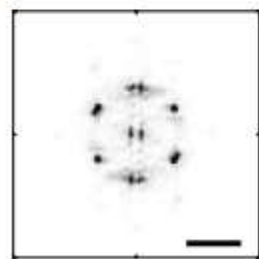
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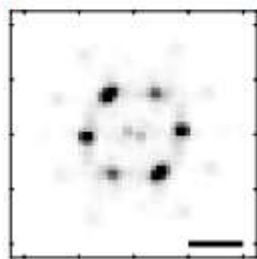
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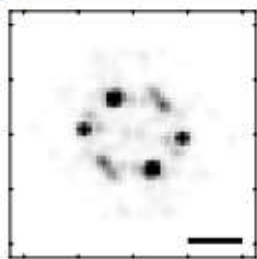
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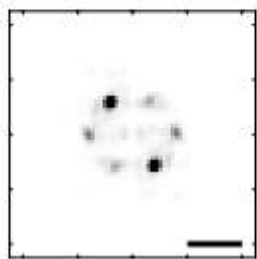
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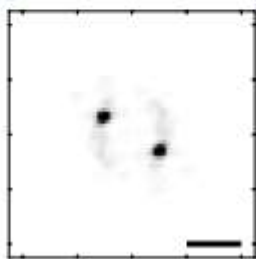
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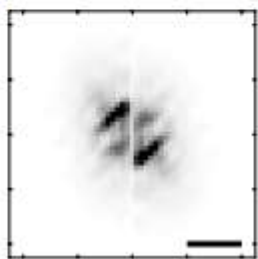
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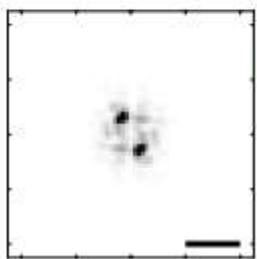
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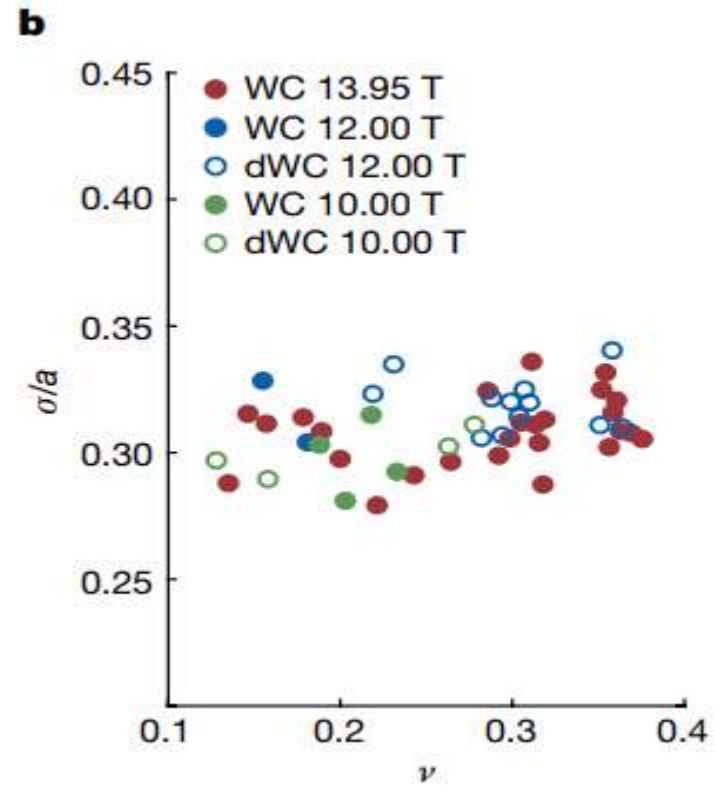
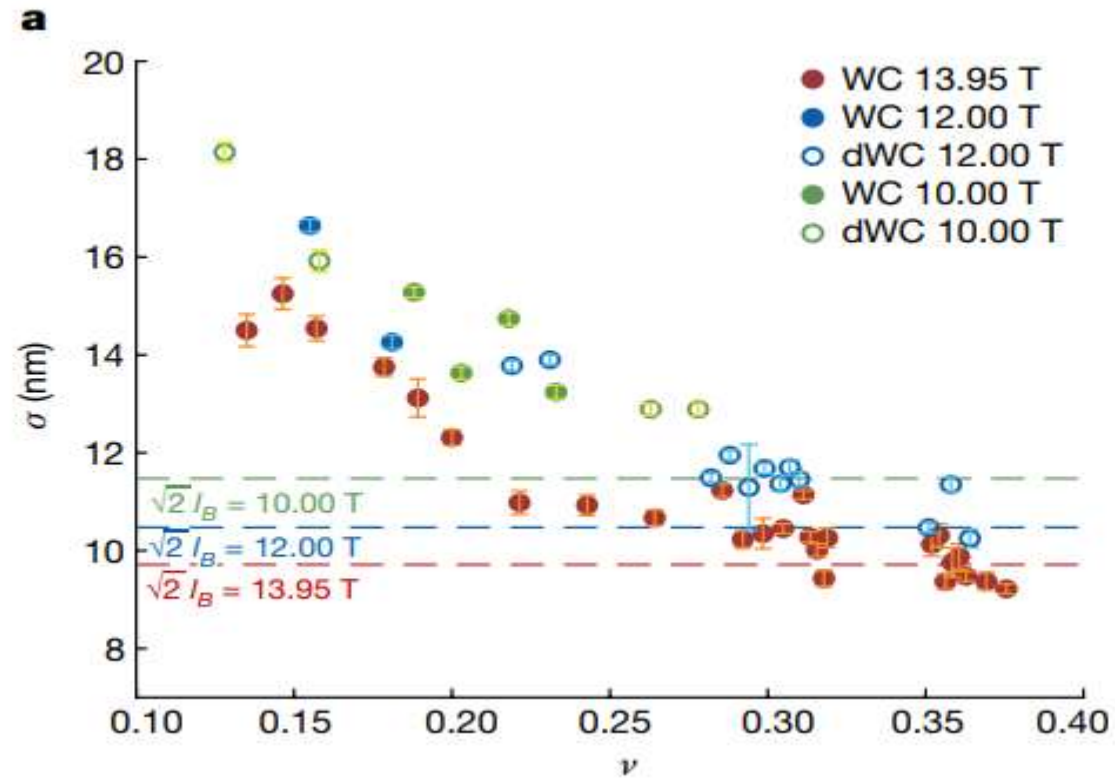


u



High
Low

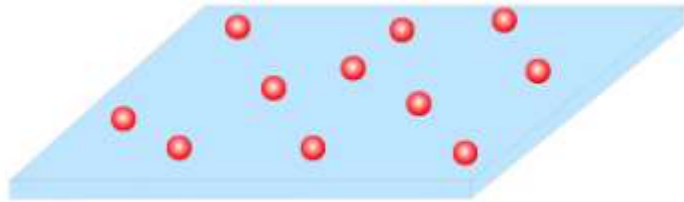
The quantum nature of the WC



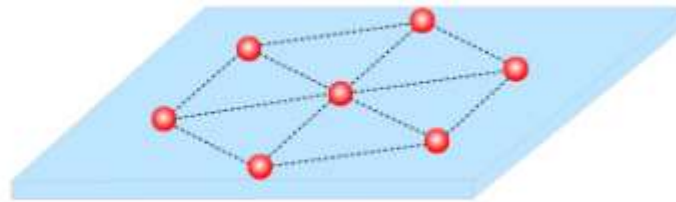
$$\mathbf{r}_i = \mathbf{R}_i + \mathbf{u}_i,$$

$$\frac{\sigma}{a} \sim 0.3.$$

Summary



Electron liquid



Wigner crystal

Experimental realization of Wigner crystal in different condensed matter systems.

References

- Wigner, E. Phys. Rev. 46, 1002–1101 (1934)
- Grimes, C. C. & Adams, G. Phys. Rev. Lett. 42, 795–798 (1979)
- Andrei, E. Y. et al. Phys. Rev. Lett. 60, 2765–2768 (1988)
- H. W. Jiang et al. Phys. Rev. Lett. 65, 633–636 (1990)
- Tsui, Y.-C. et al. Nature 628, 287–292 (2024)

Thank you for your attention!